



Blood Donation Application Using Machine Learning

Mahesh Mahajan, Makarand Shahade, Priyanka Kachave Rahul Suryawanshi*,
Prachi Patil, Shubhangi Patil

Department of Master of Computer Application, SVKM's Institute of Technology Dhule, India.

suryawanshirahul9623@gmail.com

Abstract. Blood donation is a critical aspect of health care, yet donor availability, timely supply of blood, and demand prediction continue to be problems. This paper presents a machine learning blood donation application to enhance donor-recipient interaction, predict demand for blood, and maximize blood bank management. The proposed system consists of a mobile and web application where the recipients and donors become registered, indicate their availability, and post blood request. The demand for blood is predicted based on historical records of donations, seasonal trends, and emergency demands through machine learning algorithms. The system also offers certificates and receipts as a motivation for donors. Through the use of AI/ML in healthcare, this system maximizes the effectiveness of blood donation services by minimizing shortages, and optimizing emergency response times.

Keywords: Machine Learning, Blood Donation, Realtime System, Predictive Modeling, KNN, Logistic Regression, Linear Regression, Clustering, Cloud Computing.

I.INTRODUCTION

Donating blood is really crucial for healthcare and actually saves lives in emergency situations, surgeries, and all kinds of medical procedures. But, you know, the traditional blood donation systems have a lot of issues, such as not matching the donors and recipients properly, not showing availability in real-time, and being quite poor at forecasting blood needs. Due to this, we often experience massive shortages or waste a lot of unused blood, which disrupts how well blood banks and healthcare centers function.

But, since technology is advancing, we can actually use machine learning and mobile apps to make blood donation much better and simpler. The aim of this research is to connect donors and recipients more seamlessly by providing a user-friendly, AI-based platform that not only accelerates donor matching but also encourages people to donate regularly with some awesome incentives and reminders.

2 Objective

The major objective of this study is to design an intelligent system for effective and smart blood donation management using machine learning. The system will identify and match potential donors through innovative classification methods like Decision Tree and Logistic Regression. Moreover, it aims to predict donor availability and response time through stable regression models like Random Forest, Gradient Boosting, and Linear Regression. By utilizing K-Nearest Neighbors (KNN) for donor suggestion, the system minimizes search time to make blood donations quicker. The smart framework strives to bring donors and recipients closer together, ensuring timely requirements of blood are fulfilled efficiently and depend less on conventional, human-centric processes.

3 Existing System

The traditional blood donation process is plagued by a number of drawbacks, one of which is donor-recipient matching done manually, which usually results in unsatisfactory delays in procuring donors. Manual donor eligibility also increases the likelihood of human error and invalid donor data. Absence of predictive analysis makes donor availability indefinite, forcing recipients to struggle to seek immediate assistance. Additionally, the conventional systems lack real-time tracking of location, hence it is difficult to find the location of the closest available donor in a short time. These inefficiencies necessitate the need for a sophisticated, automated system that will reduce the complexity of blood donation using data-driven strategies.

4 Proposed System

To counter the limitations of current systems, the new system incorporates machine learning methodologies for automated determination of donor eligibility, guaranteeing accuracy and effectiveness in donor selection. The application of KNN to identify the nearest donor considerably minimizes response time, facilitating rapid and effective blood donations. Predictive modeling through Random Forest, Gradient Boosting, and Linear Regression also assists in the prediction of donor availability, allowing recipients to have access to timely blood donations. Through the use of real-time tracking and smart suggestions, the system increases the overall effectiveness of the blood donation process. In addition, the use of an alert system guarantees timely notifications to both donors and recipients, promoting a proactive blood donation process and saving lives.

5 DATASET

Since real-time donor, receiver, and blood bank data are not available, a synthetic dataset was created with the help of ChatGPT. The synthetic dataset has profiles of donors and receivers with varied age, blood group, location, and medical conditions. Since the data are artificial, the structure of such data was set up to mirror real-world conditions for model training and evaluation purposes.

- Donor dataset

#	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	
1	Donor ID	Full Name	Gender	Age	Blood Gro	Phone No	Street	Add City	State	PIN Code	Country	Latitude	Longitude	Last Donation	Weight	Hemoglobin	Donation I	Availability	Hepatitis	HIV/AIDS	
2	DNR001	Mahika Di	Female	23 B-	9.18E+11	34/54	She	Jalgaon	Maharashtra	425001	India	21.0077	75.5626	6/5/2023	58	11.7	1	Available	Yes	Yes	
3	DNR002	Rhea Tara	Male	55 B	9.17E+11	53/21	Nashik	Maharashtra	422001	India	20.0059	73.7898	3/28/2023	57	15.2	0	Available	Yes	Yes	Yes	
4	DNR003	Jivika Roal	Female	47 O+	9.19E+11	H.No. 98	Nashik	Maharashtra	422001	India	20.0059	73.7898	4/19/2023	66	14.7	5	Available	Yes	Yes	Yes	
5	DNR004	Badali Bara	Male	32 A-	9.2E+11	22/63	Malegaon	Maharashtra	423203	India	20.5531	74.5288	3/15/2023	89	16	9	Available	No	Yes	No	
6	DNR005	Kabir Cont	Female	54 A-	9.19E+11	H.No. 43	Nashik	Maharashtra	422001	India	20.0059	73.7898	9/4/2023	45	15.2	2	Not Availa	Yes	Yes	Yes	
7	DNR006	Aaina Kara	Female	53 B-	9.19E+11	31/308	Malegaon	Maharashtra	423203	India	20.5531	74.5288	5/17/2024	53	13.9	2	Available	Yes	Yes	Yes	
8	DNR007	Abram Ma	Male	18 A+	9.19E+11	25	Goda F	Nashik	Maharashtra	422001	India	20.0059	73.7898	9/4/2023	58	16.3	10	Not Availa	Yes	No	No
9	DNR008	Ira Dora	Female	40 AB-	9.2E+11	692	Dhule	Maharashtra	424001	India	20.9042	74.7749	8/19/2023	68	12.9	5	Available	No	Yes	No	
10	DNR009	Devarash K	Female	37 AB+	9.18E+11	92/934	C' Nashik	Maharashtra	422001	India	20.0059	73.7898	9/9/2023	47	13	2	Not Availa	No	Yes	Yes	
11	DNR010	Kivika Ball	Male	47 O+	9.19E+11	68/716	G' Nashik	Maharashtra	422001	India	20.0059	73.7898	7/11/2024	57	13.2	8	Not Availa	No	Yes	Yes	
12	DNR011	Manjari R	Male	48 AB+	9.18E+11	951	Jalgaon	Maharashtra	425001	India	21.0077	75.5626	2/26/2023	66	14.7	8	Available	Yes	No	No	
13	DNR012	Ranbir Lok	Female	26 AB-	9.18E+11	30/739	C' Nandurbar	Maharashtra	425412	India	21.3733	74.2422	6/26/2023	66	12.1	0	Not Availa	No	Yes	Yes	
14	DNR013	Himmat R	Female	60 A-	9.18E+11	H.No. 60	Malegaon	Maharashtra	423203	India	20.5531	74.5288	2/2/2025	64	15.1	10	Available	Yes	No	No	
15	DNR014	Hiran Raju	Female	25 O+	9.2E+11	616	Vohra	Nashik	Maharashtra	422001	India	20.0059	73.7898	6/26/2023	66	13.4	4	Not Availa	No	Yes	Yes
16	DNR015	Chirag Thale	Male	55 AB-	9.17E+11	295	Chow Nandurbar	Maharashtra	425412	India	21.3733	74.2422	11/16/2023	45	13.8	2	Available	No	Yes	Yes	
17	DNR016	Baiju Halde	Female	55 A-	9.17E+11	94	Nashik	Maharashtra	422001	India	20.0059	73.7898	5/28/2023	54	12.5	10	Not Availa	No	Yes	Yes	
18	DNR017	Fayeen Gi	Male	53 B-	9.2E+11	07/65	Malegaon	Maharashtra	423203	India	20.5531	74.5288	9/13/2023	61	16.4	4	Not Availa	No	No	No	
19	DNR018	Manjari R	Female	26 O+	9.18E+11	88/925	Sa Dhule	Maharashtra	424001	India	20.9042	74.7749	11/12/2023	72	11.9	4	Available	Yes	Yes	Yes	
20	DNR019	Mishra Bor	Female	45 AB+	9.18E+11	56/128	Nashik	Maharashtra	422001	India	20.0059	73.7898	2/6/2023	68	13.8	0	Not Availa	Yes	Yes	Yes	
21	DNR020	Nirvi Dube	Female	59 A-	9.2E+11	H.No. 17	Jalgaon	Maharashtra	425001	India	21.0077	75.5626	7/16/2023	46	14.1	6	Available	Yes	Yes	Yes	
22	DNR021	Ayaz Lal	Female	19 A-	9.19E+11	H.No. 36	Malegaon	Maharashtra	423203	India	20.5531	74.5288	1/2/2024	73	12.2	10	Not Availa	Yes	No	No	
23	DNR022	Elaa Khatri	Male	53 B+	9.17E+11	833	Jalgaon	Maharashtra	425001	India	21.0077	75.5626	8/5/2023	84	15.2	2	Available	Yes	No	No	
24	DNR023	Kanar Gon	Male	37 B+	9.18E+11	16/224	La Jalgaon	Maharashtra	425001	India	21.0077	75.5626	12/29/2024	74	13.2	8	Not Availa	No	Yes	Yes	
25	DNR024	Fayeen Gi	Female	59 AB-	9.19E+11	68	Dewy 2	Malegaon	Maharashtra	423203	India	20.5531	74.5288	9/13/2023	49	14.3	0	Available	No	No	No
26	DNR025	Parineez R	Male	52 B+	9.18E+11	H.No. 13	Malegaon	Maharashtra	423203	India	20.5531	74.5288	9/25/2024	59	13.6	3	Not Availa	No	Yes	Yes	
27	DNR026	Lagan Siva	Female	27 AB-	9.18E+11	160	Nashik	Maharashtra	422001	India	20.0059	73.7898	10/5/2023	57	12	7	Not Availa	Yes	Yes	Yes	
28	DNR027	Mishri Kha	Female	31 O-	9.18E+11	99/53	Nashik	Maharashtra	422001	India	20.0059	73.7898	12/14/2023	70	14.6	9	Not Availa	No	No	No	
29	DNR028	Eliza B	Female	47 B-	9.19E+11	H.No. 880	Malegaon	Maharashtra	423203	India	20.5531	74.5288	10/13/2024	70	16.1	3	Not Availa	No	No	No	

Fig.1 Donor Dataset

- Blood Receiver Dataset

#	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	Receiver ID	Full Name	Gender	Weight	Age	Blood Gro	Phone No	Email	Street	Add City	State/Pro	Zip/Postal	Country	Latitude	Longitude	Hospital	Urgency	U Medical	C Units Req	Required Date	Donation Status	
2	1	Rehan G	Male	58	65 B-	1.59E+08	Intikash81	81	Chahan Mohadi	Maharashtra	424301	India	21.34896	74.78835	Aditya Hos	Medium	Cancer	Tric	5	2/15/2025	Fulfilled	
3	2	Priya Nari	Male	57	53 O+	9.14E+08	bondarchat	46/880	Bi Dondachi	Maharashtra	425408	India	20.57468	74.9383	Janta Hos	Medium	Truma	5	2/14/2025	Pending		
4	3	Advesta Pi	Female	66	22 O-	3.92E+09	jaosbarha	H.No. 23	(Dondachi	Maharashtra	425408	India	20.73385	74.82779	Aditya Hos	Critical	Cancer	Tric	1	2/22/2025	Pending	
5	4	Zehaan Sh	Female	89	55 A-	7.48E+08	larampana	58	Sarker Mohadi	Maharashtra	424301	India	21.18372	74.75112	Lamni Hos	Medium	Blood Loss	5	2/17/2025	Fulfilled		
6	5	Fredrick F	Female	45	27 AB-	9.43E+08	sahharwal	03/29	Tat Dondachi	Maharashtra	425408	India	21.16996	74.84801	Janta Hos	High	Blood Loss	1	2/25/2025	Pending		
7	6	Balshah L	Male	53	23 A-	9.1E+08	epanibh	69/69	Tat Dondachi	Maharashtra	425408	India	20.67619	74.94021	Janta Hos	Medium	Chronic Cr	4	2/12/2025	Pending		
8	7	Gautam S	Female	58	49 O+	9.16E+11	sakharwal	569	Wadi Sakhi	Maharashtra	424303	India	21.42825	75.25149	City Hospi	Critical	Trauma	4	2/19/2025	Fulfilled		
9	8	Yashasi G	Male	68	50 A+	9.14E+11	yatan540	13	Sampa Shirpur	Maharashtra	425405	India	21.33889	75.43249	City Hospi	Critical	Severe Anx	4	2/12/2025	Fulfilled		
10	9	Mughla A	Male	47	41 AB-	2.08E+09	karlayib	415	Bau Dhule	Maharashtra	424001	India	21.39549	74.85044	Sadha Hos	Medium	Chronic Cr	2	2/10/2025	Pending		
11	10	Chandani	Female	57	23 A+	9.12E+11	mlyengar	H.No. 86	B' Shirpur	Maharashtra	425405	India	21.15738	75.24112	Aditya Hos	High	Cancer	Tric	5	2/20/2025	Fulfilled	
12	11	Adya Om	Female	66	61 AB-	3.98E+08	sankarpar	86	Savini Nandurbar	Maharashtra	425412	India	21.10315	74.75371	City Hospi	Critical	Blood Loss	2	2/22/2025	Fulfilled		
13	12	Niharika L	Male	66	64 A-	7.16E+08	fradepur	H.No. 206	Sakhi	Maharashtra	424303	India	20.63489	75.36576	Lamni Hos	High	Trauma	1	2/14/2025	Pending		
14	13	Bhavana K	Male	64	45 AB+	2.47E+08	gureshdas	43/991	M Dondachi	Maharashtra	425408	India	20.86071	75.37861	Lamni Hos	High	Blood Loss	1	2/14/2025	Pending		
15	14	Upadhyai I	Male	66	37 O+	9.20E+11	yadan506	88/095	Ra Dondachi	Maharashtra	425408	India	20.79105	75.15135	City Hospi	Critical	Trauma	1	2/15/2025	Pending		
16	15	Yashika R	Female	45	30 O+	8.03E+08	ishanilalia	H.No. 672	Mohadi	Maharashtra	424301	India	20.83427	75.08399	Janta Hos	Critical	Trauma	1	2/12/2025	Fulfilled		
17	16	Samar Nar	Male	54	62 AB+	2.84E+08	amshilake	H.No. 142	Sindhedha	Maharashtra	424304	India	21.38842	74.64126	City Hospi	Critical	Cancer	Tric	4	2/13/2025	Pending	
18	17	Arjan Cho	Female	61	51 A-	8.47E+08	varshata	77/832	M Nandurbar	Maharashtra	425412	India	21.36554	74.66504	Dhule Gen	Critical	Trauma	4	2/16/2025	Pending		
19	18	Omkar H	Male	72	61 A+	9.19E+11	agastyasi	H.No. 565	Shirpur	Maharashtra	425405	India	20.94228	74.74246	City Hospi	Critical	Severe Anx	5	2/22/2025	Fulfilled		
20	19	Gaurang H	Female	68	29 A-	5.85E+08	hailuwalla	34/49	Nay Dondachi	Maharashtra	425408	India	20.81283	75.19527	Lamni Hos	Critical	Trauma	4	2/20/2025	Fulfilled		
21	20	Meghana I	Female	46	43 AB-	3.26E+08	trivanarati	22/37	Gre Sakhi	Maharashtra	424303	India	20.78179	74.51975	Aditya Hos	High	Blood Loss	3	2/15/2025	Pending		
22	21	Chasnum	Female	73	33 O+	6.02E+08	anuramk	H.No. 885	Sakhi	Maharashtra	424303	India	20.5861	75.04512	City Hospi	Medium	Cancer	Tric	2	2/12/2025	Pending	
23	22	Jagat Sahni	Male	84	24 A+	9.16E+11	esemjain	H.No. 794	Dondachi	Maharashtra	425408	India	20.67966	74.62438	Aditya Hos	High	Cancer	Tric	2	2/18/2025	Pending	
24	23	Lohit Dasi	Female	74	22 O+	1.94E+09	abhinmay	22/186	Ta Sakhi	Maharashtra	424303	India	21.13884	75.27855	Lamni Hos	Medium	Major Surg	1	2/12/2025	Pending		
25	24	Varish Sar	Male	49	59 A+	6.53E+08	mamajharia	H.No. 327	Dhule	Maharashtra	424001	India	21.01671	74.86889	City Hospi	High	Major Surg	1	2/10/2025	Pending		
26	25	Sita Dasi	Female	59	24 B-	2.42E+08	mangalghat	H.No. 35	Sindhedha	Maharashtra	424304	India	21.00753	74.83075	Lamni Hos	High	Trauma	2	2/24/2025	Pending		
27	26	Ujaya Sain	Female	57	39 O+	7.44E+08	anuratharaj	81/81	Gre Sakhi	Maharashtra	424303	India	21.41973	75.44061	Lamni Hos	Critical	Chronic Cr	2	2/12/2025	Pending		
28	27	Ancora Des	Female	70	35 AB-	7.59E+08	shrayasid	H.No. 21	J' Dhule	Maharashtra	424001	India	21.3819	74.86613	Aditya Hos	Critical	Chronic Cr	1	2/24/2025	Fulfilled		
29	28	Shikha P	Female	70	62 B-	6.33E+08	anuramk	56/618	Gre Sakhi	Maharashtra	424303	India	20.47846	74.86213	Lamni Hos	Critical	Blood Loss	3	2/12/2025	Pending		

Fig.2 Blood Receiver Dataset

- Blood Bank Dataset

	A	B	C	D	E
1	Blood Bank ID	Name of Blood Bank	Contact No.	Address	Available Blood Units
2	BB001	Dhule Civil Hospital Blood Bank	02562-234567	Near Civil Hospital, Dhule	O+: 10, A+: 8, B+: 6, AB+: 2
3	BB002	Red Cross Blood Bank	02562-245678	MG Road, Dhule	O+: 7, A+: 5, B+: 4, AB+: 3
4	BB003	Sakri Blood Center	02563-225432	Sakri Main Road, Sakri	O+: 5, A+: 6, B+: 3, AB+: 1
5	BB004	Shirpur Blood Bank	02563-235678	Near Bus Stand, Shirpur	O+: 4, A+: 5, B+: 7, AB+: 2
6	BB005	Sindkheda Health Blood Bank	02563-215432	Civil Hospital, Sindkheda	O+: 6, A+: 3, B+: 5, AB+: 1

Fig.3 Blood Bank Dataset

6 Machine Learning Algorithm:

1.Logistic Regression:

Predicts the probability of a donor donating in the future given the past frequency of donation, age, and health. It is useful in sending timely reminders to potential donors.

2.Decision Tree & Random Forest:

Classifies donors as eligible or ineligible based on age, weight, hemoglobin levels, and medical conditions. Random Forest improves accuracy by using multiple decision trees for robust predictions.

3.K-Nearest Neighbors (KNN):

Matches donors to recipients by proximity, blood type compatibility, and level of urgency. Gives way to emergency cases where quick identification of the closest available donor is needed.

Structure of Blood Donation Application

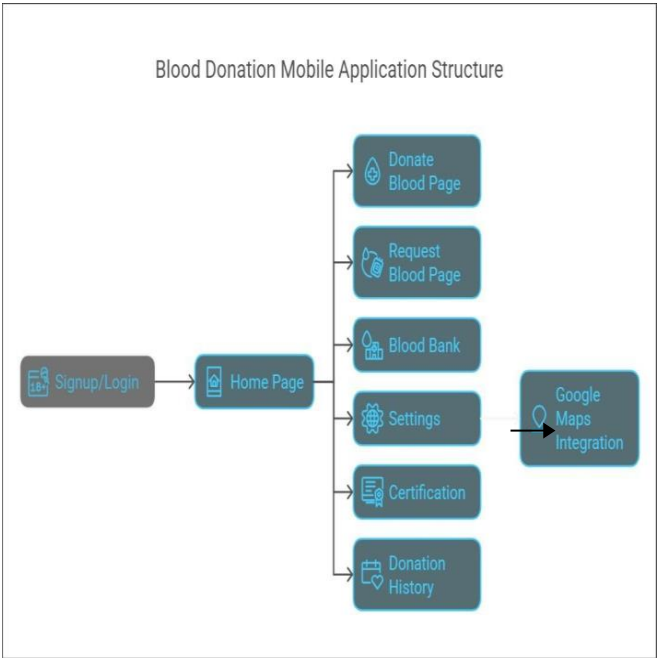


Fig.4.Blood Donation Mobile Application Structure

RESULTS

Four models were developed using Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), and K-Nearest Neighbors (KNN) to improve the efficiency of blood donation systems: each model was trained on different aspects of donor eligibility classification and matching donor to recipient. the comparison and evaluation performance of such models based on measures of accuracy, precision, recall, and F1-score.

Table 1 : accuracy

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	80.28	80.29	80.28	80.28
Decision Tree	79.28	81.31	79.28	79.04
Random Forest	83.47	83.51	83.47	83.47
K-NN	85.66	86.44	85.66	85.54

CONCLUSION AND FUTURE TRENDS

This paper focuses on improving the blood donation system through the use of machine learning techniques. The proposed system makes use of Logistic Regression, Decision Tree, Random Forest, and K-Nearest Neighbors (KNN) to make enhancements about donor selection and emergency response. The results clearly indicate that under standard conditions, the system gives good predictions and hence behaves reliably. However, there are some challenges with misclassifications, processing times consuming abundant datasets, and limitations in real-time tracking of donors. A dataset was used to test performance, ensuring the model was reliable in real-world applications. These improvements are expected to enhance the donor availability prediction capability and overall operational efficiency in managing blood donations.

REFERENCES

1. Paper, I. J. R. A. S. E. T. (2023). Creating and deploying an online blood bank administration system to facilitate effective blood donation and distribution. *Applied Science & Engineering Technology Research International Journal (IJRASET)*, 11(IV), 1830–1834.
2. Orlando Iparraguirre-Villanueva, Fernando Sierra-Liñan, and Michael Cabanillas-Carbonell. (2022). A mobile application that uses location to find blood donors. *Advanced Computer Science and Applications International Journal*, 13. doi:10.14569/IJACSA.2022.0130418.
3. P. L. Wijayathilaka, P. H. Pahala Gamage, K. H. B. De Silva, A. P. P. S. Athukorala, K. A. D. C. P. Kahandawaarachchi, and K. N. Pulasinghe, “Secured, intelligent blood and organ donation management system – “Lifeshare””/0. ICAC 2020 - 2nd International Conference on Advancements in Computing, Proceedings, pp. 374– 379, Dec. 2020, doi:10.1109/ICAC51239.2020.9357211.

4. L. Dutta, G. Maji, P. Ghosh, and S. Sen, “An integrated blood donation campaign management system,” *Advances in Intelligent Systems and Computing*, vol. 812, pp. 133–143, 2019, doi:10.1007/9789811315404_14.
5. R. S. Ali, T. F. Hafez, A. B. Ali, and N. AbdAlsabour, “Blood bag: A web application to manage all blood donation and transfusion processes,” *Proceedings of the 2017 International Conference on Wireless Communications, Signal Processing and Networking, WiSPNET 2017*, vol. 2018-January, pp. 2125–2130, Feb. 2018, doi: 10.1109/WISPNET.2017.8300136.
6. M. Alicia et al., “YAWARweb: Pilot study about the usage of a web service to raise awareness of blood donation campaigns on university campuses in Lima, Peru,” *International Journal of Advanced Computer Science and Applications*, vol. 10, no. 7, pp. 402–407, 2019, doi: 10.14569/IJACSA.2019.0100755.
7. M. Nabil, R. Ihab, H. elMasry, S. Said, and S. Youssef, “A Web-based blood donation and Medical Monitoring System

Open Access This chapter is licensed under the terms of the Creative Commons Attribution-NonCommercial 4.0 International License (<http://creativecommons.org/licenses/by-nc/4.0/>), which permits any noncommercial use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license and indicate if changes were made.

The images or other third party material in this chapter are included in the chapter's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the chapter's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder.

